

MA.912.AR.1.2

### **Benchmark**

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**MA.912.AR.1.2** Rearrange equations or formulas to isolate a quantity of interest.

*Algebra I Example:* The Ideal Gas Law  $PV = nRT$  can be rearranged as  $T = \frac{PV}{nR}$  to isolate temperature as the quantity of interest.

*Example:* Given the Compound Interest formula  $A = P(1 + \frac{r}{n})^{nt}$ , solve for  $P$ .

*Mathematics for Data and Financial Literacy Honors Example:* Given the Compound Interest formula  $A = P(1 + \frac{r}{n})^{nt}$ , solve for  $t$ .

**Benchmark Clarifications:**

*Clarification 1:* Instruction includes using formulas for temperature, perimeter, area and volume; using equations for linear (standard, slope-intercept and point-slope forms) and quadratic (standard, factored and vertex forms) functions.

*Clarification 2:* Within the Mathematics for Data and Financial Literacy course, problem types focus on money and business.

## Connecting Benchmarks/Horizontal Alignment

- MA.912.NSO.1.2
- MA.912.AR.2.2, MA.912.AR.2.5, MA.912.AR.2.6
- MA.912.AR.3.1, MA.912.AR.3.6, MA.912.AR.3.7, MA.912.AR.3.8
- MA.912.AR.4.1
- MA.912.AR.5.3, MA.912.AR.5.6
- MA.912.FL.3.2

## Terms from the K-12 Glossary

- Equation

## Vertical Alignment

### Previous Benchmarks

- MA.8.AR.2
- MA.8.GR.1

### Next Benchmarks

- MA.912.AR.5.5, MA.912.AR.5.9
- MA.912.AR.8.2
- MA.912.T.3.2

## Purpose and Instructional Strategies

In grade 8, students isolated variables in one-variable linear equations and one-variable quadratic equations in the form  $x^2 = p$  and  $x^3 = q$ . In Algebra I, students isolate a variable or quantity of interest in equations and formulas. Equations and variables will focus on linear, absolute value and quadratic in Algebra I. In later courses, students will highlight a variable or quantity of interest for other types of equations and formulas, including exponential, logarithmic and trigonometric.

- Instruction includes making connections to inverse arithmetic operations (refer to Appendix D) and solving one-variable equations.
- Instruction includes justifying each step while rearranging an equation or formula.
  - For example, when rearranging  $A = P \left(1 + \frac{r}{n}\right)^{nt}$  for  $P$ , it may be helpful for students to highlight the quantity of interest with a highlighter, so students remain focused on that quantity for isolation purposes. It may also be helpful for students to identify factors, or other parts of the equations.

## Common Misconceptions or Errors

- Students may not have mastered the inverse arithmetic operations.
- Students may be frustrated because they are not arriving at a numerical value as their solution. Remind students that they are rearranging variables that can be later evaluated to a numerical value.
- Having multiple variables and no values may confuse students and make it difficult for them to see the connections between rearranging a formula and solving a one-variable equation.

## Strategies to Support Tiered Instruction

- Instruction includes doing a side-by-side comparison of solving a multistep equation with rearranging equations and formulas. The teacher should allow students time to understand that the steps in solving both equations are the same.
  - For example, solve both equations and note the similarities in solving both types of equations.

| Solving One-Variable Equations                                                                                                                                                                                                                                                                                                                                                        | Rearranging Equations/Formulas                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Determine the height, <math>h</math>, in the formula <math>SA = 2B + Ph</math> if the surface area (<math>SA</math>) is 537 units squared, the area of the base (<math>B</math>) is 112 units squared and the perimeter of the base (<math>P</math>) is 25 units.</p> $537 = 2(112) + 25h$ $537 - 224 = 224 + 25h - 224$ $313 = 25h$ $\frac{313}{25} = \frac{25h}{25}$ $12.52 = h$ | <p>Isolate height, <math>h</math>, in the formula <math>SA = 2B + Ph</math>.</p> $SA = 2B + Ph$ $SA - 2B = 2B + Ph - 2B$ $SA - 2B = Ph$ $\frac{SA - 2B}{P} = \frac{Ph}{P}$ $\frac{SA - 2B}{P} = h$ |

- Teacher provides a chart for students to use as a study guide or to copy in their interactive notebook.
  - For example, inverse operations chart below.

| Inverse Operations Chart |   |                           |
|--------------------------|---|---------------------------|
| Addition<br>+            | ↔ | Subtraction<br>−          |
| Multiplication<br>×      | ↔ | Division<br>÷             |
| Square<br>$x^2$          | ↔ | Square Root<br>$\sqrt{x}$ |

## Instructional Tasks

### Instructional Task 1 (MTR.4.1, MTR.5.1)

Part A. Given the equation  $ax^2 + bx + c = 0$ , solve for  $x$ .

Part B. Share your strategy with a partner. What do you notice about the new equation(s)?

### Instructional Task 2 (MTR.4.1, MTR.5.1)

Part A. Given the equation  $Ax + By = C$ , solve for  $B$ .

Part B. Given the equation  $7x - 6y = 24$ , determine the  $x$ - and  $y$ -intercepts.

Part C. What do you notice between Part A and Part B?

## Instructional Items

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### *Instructional Item 1*

Solve for  $x$  in the equation  $3x + y = 5x - xy$ .

### *Instructional Item 2*

The formula  $d = \frac{v_o + v_t}{2} t$  relating to the translational of motion, where  $d$  represents distance,  $v_o$  represents initial velocity,  $v_t$  represents final velocity, and  $t$  represents time. Rearrange the formula to isolate final velocity.

### *Instructional Item 3*

The area  $A$  of a sector of a circle with radius  $r$  and angle-measure  $S$  (in degrees) is given by  $A = \frac{\pi r^2 S}{360}$ , solve for the radius  $r$ .

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*\*The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*